

Pandemic-Period Comparison

Psycholinguistic markers pre / during / post pandemic
after Yahya & Abdul Rahim (2023), with per-user rank-based statistics

Dataset: Combined Dataset

Section 1 — Diagnostics

Period Definitions

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pre      : PostDate < 2020-03-11
during   : 2020-03-11 ≤ PostDate < 2022-03-23
post     : PostDate ≥ 2022-03-23
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Cutoff = WHO pandemic declaration (2020-03-11).

End = end of most Dutch COVID restrictions (2022-03-23).

Dataset Variant × Pandemic Period — messages and unique users
⚠ **Period effects are confounded with dataset variant (old export ≈ pre+during, new export ≈ post); yellow = near-empty cell**

variant	period	messages	users
old	pre	4505	169
old	during	11102	291
old	post	2943	96
new_only	pre	94	15
new_only	during	318	23
new_only	post	25092	520
combined	pre	4910	178
combined	during	12036	307
combined	post	28235	604

Users by Number of Periods Appeared In
Users in >1 period violate between-group independence — see the single-period sensitivity analysis in Section 4

periods_appeared_in	n_users	pct_users
1	816	86.8
2	99	10.5
3	25	2.7

Section 2 — Dictionary Introspection

Yahya & Abdul Rahim (2023) §2.3 Categories vs Available Dictionaries
Only available categories are analyzed; '—' = genuinely absent from that source (omission, not a bug)

yahya_group	liwc2015	liwc22	dic_crosscheck
pronouns	i, ipron, ppron, pronoun, shehe, they, we, you	i, ipron, ppron, pronoun, shehe, they, we, you	i, ipron, ppron, pronoun, shehe, they, we, you
function_words	article, auxverb, conj, function, negate, number, prep, quant	article, auxverb, conj, function, negate, number, prep, quant	article, auxverb, conj, function, negate, number, prep, quant
other_grammar	adj, adverb, compare, interrog, verb	adj, adverb, compare, interrog, verb	adj, adverb, compare, interrog, verb
tense	focusfuture, focuspast, focuspresent	focusfuture, focuspast, focuspresent	focusfuture, focuspast, focuspresent
emotion	affect, anger, anx, negemo, posemo, sad	affect, anger, anx, negemo, posemo, sad	affect, anger, anx, negemo, posemo, sad
informal	assent, filler, informal, netspeak, nonflu, swear	assent, filler, informal, netspeak, nonflu, swear	assent, filler, informal, netspeak, nonflu, swear
summary	—	—	—
absolutist	absolutist	—	—

Section 3 — Main Results (all users)

Kruskal-Wallis Omnibus per Feature (per-user rates)
effect_size = epsilon-squared; p_bh = BH over pooled omnibus + post-hoc family; yellow = significant

feature	n_pre	n_during	n_post	comparison	statistic	p_raw	effect_size	note	p_bh	significant
conj_liwc22	178	307	604	omnibus	7.0511	0.029436	0.0065		0.224076	False
anger_liwc2015	178	307	604	omnibus	7.0534	0.029402	0.0065		0.224076	False
shehe_liwc22	178	307	604	omnibus	7.2031	0.027282	0.0066		0.224076	False
pronoun_liwc22	178	307	604	omnibus	8.6231	0.013413	0.0079		0.224076	False
anger_liwc22	178	307	604	omnibus	7.0119	0.030018	0.0064		0.224076	False
swear_liwc2015	178	307	604	omnibus	7.3713	0.025081	0.0068		0.224076	False
shehe_liwc2015	178	307	604	omnibus	7.1465	0.028064	0.0066		0.224076	False
conj_liwc2015	178	307	604	omnibus	6.9673	0.030695	0.0064		0.224076	False
swear_liwc22	178	307	604	omnibus	7.4067	0.02464	0.0068		0.224076	False
pronoun_liwc2015	178	307	604	omnibus	9.0926	0.010606	0.0084		0.224076	False
prep_liwc2015	178	307	604	omnibus	6.3734	0.041308	0.0059		0.23096	False
prep_liwc22	178	307	604	omnibus	6.4551	0.039655	0.0059		0.23096	False
interrog_liwc2015	178	307	604	omnibus	6.2908	0.04305	0.0058		0.23096	False
interrog_liwc22	178	307	604	omnibus	6.2338	0.044294	0.0057		0.23096	False
ppron_liwc2015	178	307	604	omnibus	5.7635	0.056035	0.0053		0.272706	False
ppron_liwc22	178	307	604	omnibus	5.5873	0.061198	0.0051		0.279217	False
posemo_liwc22	178	307	604	omnibus	5.226	0.073313	0.0048		0.306788	False
posemo_liwc2015	178	307	604	omnibus	5.1634	0.075646	0.0047		0.306788	False
they_liwc2015	178	307	604	omnibus	4.6873	0.095975	0.0043		0.350308	False
they_liwc22	178	307	604	omnibus	4.7452	0.093236	0.0044		0.350308	False
function_liwc22	178	307	604	omnibus	4.3312	0.114679	0.004		0.398645	False
function_liwc2015	178	307	604	omnibus	4.0845	0.129734	0.0038		0.430481	False
anx_liwc22	178	307	604	omnibus	3.55	0.169484	0.0033		0.503942	False
informal_liwc22	178	307	604	omnibus	3.633	0.162595	0.0033		0.503942	False
anx_liwc2015	178	307	604	omnibus	3.5138	0.172583	0.0032		0.503942	False
negemo_liwc22	178	307	604	omnibus	3.3205	0.190094	0.0031		0.518384	False
informal_liwc2015	178	307	604	omnibus	3.3033	0.191731	0.003		0.518384	False
negemo_liwc2015	178	307	604	omnibus	3.1895	0.202962	0.0029		0.51949	False
netspeak_liwc2015	178	307	604	omnibus	3.1561	0.206373	0.0029		0.51949	False
assent_liwc22	178	307	604	omnibus	2.8902	0.235728	0.0027		0.537753	False
assent_liwc2015	178	307	604	omnibus	2.9068	0.233773	0.0027		0.537753	False
netspeak_liwc22	178	307	604	omnibus	2.8942	0.23525	0.0027		0.537753	False
quant_liwc22	178	307	604	omnibus	1.9081	0.385184	0.0018		0.666614	False
adverb_liwc22	178	307	604	omnibus	1.7295	0.421165	0.0016		0.666614	False
i_liwc22	178	307	604	omnibus	1.6605	0.435948	0.0015		0.666614	False
auxverb_liwc22	178	307	604	omnibus	2.0638	0.356322	0.0019		0.666614	False
article_liwc22	178	307	604	omnibus	1.5434	0.462236	0.0014		0.666614	False
adverb_liwc2015	178	307	604	omnibus	1.7753	0.411617	0.0016		0.666614	False
ipron_liwc22	178	307	604	omnibus	1.6195	0.444961	0.0015		0.666614	False
verb_liwc2015	178	307	604	omnibus	1.8679	0.392992	0.0017		0.666614	False
sad_liwc22	178	307	604	omnibus	2.2337	0.32731	0.0021		0.666614	False
focuspresent_liwc2015	178	307	604	omnibus	1.5217	0.467266	0.0014		0.666614	False
auxverb_liwc2015	178	307	604	omnibus	1.9395	0.379173	0.0018		0.666614	False
sad_liwc2015	178	307	604	omnibus	2.2995	0.316719	0.0021		0.666614	False
quant_liwc2015	178	307	604	omnibus	1.9175	0.383371	0.0018		0.666614	False
verb_liwc22	178	307	604	omnibus	2.0223	0.363802	0.0019		0.666614	False
focuspast_liwc22	178	307	604	omnibus	1.9049	0.385798	0.0018		0.666614	False
focuspresent_liwc22	178	307	604	omnibus	1.7793	0.410797	0.0016		0.666614	False
ipron_liwc2015	178	307	604	omnibus	1.5088	0.470301	0.0014		0.666614	False
focuspast_liwc2015	178	307	604	omnibus	1.9709	0.373276	0.0018		0.666614	False
i_liwc2015	178	307	604	omnibus	1.7141	0.424422	0.0016		0.666614	False
absolutist_liwc2015	178	307	604	omnibus	1.4895	0.474848	0.0014		0.666614	False
article_liwc2015	178	307	604	omnibus	1.4049	0.495362	0.0013		0.68229	False
focusfuture_liwc22	178	307	604	omnibus	1.2197	0.543439	0.0011		0.690846	False
filler_liwc22	178	307	604	omnibus	1.212	0.545532	0.0011		0.690846	False
you_liwc22	178	307	604	omnibus	1.2866	0.525566	0.0012		0.690846	False
focusfuture_liwc2015	178	307	604	omnibus	1.1997	0.548891	0.0011		0.690846	False
you_liwc2015	178	307	604	omnibus	1.2262	0.541664	0.0011		0.690846	False
number_liwc22	178	307	604	omnibus	1.1291	0.568627	0.001		0.703555	False
negate_liwc22	178	307	604	omnibus	0.9892	0.609826	0.0009		0.718021	False
filler_liwc2015	178	307	604	omnibus	1.0208	0.600258	0.0009		0.718021	False
negate_liwc2015	178	307	604	omnibus	1.0117	0.602981	0.0009		0.718021	False
compare_liwc2015	178	307	604	omnibus	0.9411	0.624646	0.0009		0.719842	False
compare_liwc22	178	307	604	omnibus	0.9206	0.631094	0.0008		0.719842	False
number_liwc2015	178	307	604	omnibus	0.6825	0.710872	0.0006		0.798364	False
adj_liwc22	178	307	604	omnibus	0.5714	0.7515	0.0005		0.825417	False
adj_liwc2015	178	307	604	omnibus	0.5553	0.757574	0.0005		0.825417	False
nonflu_liwc2015	178	307	604	omnibus	0.364	0.8336	0.0003		0.894894	False
nonflu_liwc22	178	307	604	omnibus	0.3185	0.852768	0.0003		0.902204	False
we_liwc22	178	307	604	omnibus	0.1036	0.949505	0.0001		0.976252	False
we_liwc2015	178	307	604	omnibus	0.1198	0.941854	0.0001		0.976252	False
affect_liwc22	178	307	604	omnibus	0.0157	0.992205	0.0		0.994621	False
affect_liwc2015	178	307	604	omnibus	0.0108	0.994621	0.0		0.994621	False

No features survived omnibus BH correction — no post-hoc tests run

Section 4 — Sensitivity: Single-Period Users Only

Same Design, Restricted to Users Appearing in Exactly One Period
Removes the between-group dependence of multi-period users; compare significance patterns with Section 3

feature	n_pre	n_during	n_post	comparison	statistic	p_raw	effect_size	note	p_bh	significant
netspeak_liwc22	92.0	185.0	539.0	omnibus	13.4567	0.001197	0.0165		0.01361	True
netspeak_liwc2015	92.0	185.0	539.0	omnibus	14.0036	0.00091	0.0172		0.01361	True
pronoun_liwc22	92.0	185.0	539.0	omnibus	13.613	0.001107	0.0167		0.01361	True
pronoun_liwc2015	92.0	185.0	539.0	omnibus	13.9272	0.000946	0.0171		0.01361	True
ppron_liwc2015	92.0	185.0	539.0	omnibus	11.9042	0.0026	0.0146		0.022876	True
ppron_liwc22	92.0	185.0	539.0	omnibus	11.1568	0.003779	0.0137		0.028654	True
informal_liwc22	92.0	185.0	539.0	omnibus	9.1848	0.010128	0.0113		0.058764	False
informal_liwc2015	92.0	185.0	539.0	omnibus	8.3553	0.015335	0.0103		0.077526	False
swear_liwc22	92.0	185.0	539.0	omnibus	7.9828	0.018474	0.0098		0.083829	False
swear_liwc2015	92.0	185.0	539.0	omnibus	7.957	0.018714	0.0098		0.083829	False
i_liwc2015	92.0	185.0	539.0	omnibus	7.8906	0.019345	0.0097		0.083829	False
i_liwc22	92.0	185.0	539.0	omnibus	7.4811	0.023741	0.0092		0.096282	False
function_liwc22	92.0	185.0	539.0	omnibus	6.5625	0.037582	0.0081		0.136798	False
function_liwc2015	92.0	185.0	539.0	omnibus	6.2576	0.043771	0.0077		0.153199	False
absolutist_liwc2015	92.0	185.0	539.0	omnibus	5.7103	0.057547	0.007		0.193954	False
conj_liwc2015	92.0	185.0	539.0	omnibus	5.1331	0.076802	0.0063		0.230457	False
conj_liwc22	92.0	185.0	539.0	omnibus	5.2697	0.071731	0.0065		0.230457	False
interrog_liwc2015	92.0	185.0	539.0	omnibus	5.0891	0.078507	0.0062		0.230457	False
interrog_liwc22	92.0	185.0	539.0	omnibus	5.1009	0.078048	0.0063		0.230457	False
ipron_liwc22	92.0	185.0	539.0	omnibus	4.158	0.125054	0.0051		0.307	False
article_liwc22	92.0	185.0	539.0	omnibus	4.1655	0.124589	0.0051		0.307	False
adj_liwc22	92.0	185.0	539.0	omnibus	4.1883	0.123174	0.0051		0.307	False
shehe_liwc2015	92.0	185.0	539.0	omnibus	4.1506	0.125521	0.0051		0.307	False
shehe_liwc22	92.0	185.0	539.0	omnibus	4.2427	0.119867	0.0052		0.307	False
adj_liwc2015	92.0	185.0	539.0	omnibus	4.2649	0.118549	0.0052		0.307	False
filler_liwc22	92.0	185.0	539.0	omnibus	4.1084	0.128198	0.005		0.307	False
article_liwc2015	92.0	185.0	539.0	omnibus	4.0198	0.134002	0.0049		0.312672	False
assent_liwc22	92.0	185.0	539.0	omnibus	3.8613	0.145057	0.0047		0.314291	False
assent_liwc2015	92.0	185.0	539.0	omnibus	3.8874	0.143176	0.0048		0.314291	False
ipron_liwc2015	92.0	185.0	539.0	omnibus	3.8776	0.143875	0.0048		0.314291	False
filler_liwc2015	92.0	185.0	539.0	omnibus	3.4577	0.177492	0.0042		0.375623	False
number_liwc22	92.0	185.0	539.0	omnibus	3.1347	0.208601	0.0038		0.421838	False
anger_liwc2015	92.0	185.0	539.0	omnibus	2.7948	0.247244	0.0034		0.46299	False
anger_liwc22	92.0	185.0	539.0	omnibus	2.7782	0.249302	0.0034		0.46299	False
number_liwc2015	92.0	185.0	539.0	omnibus	2.8368	0.242104	0.0035		0.46299	False
prep_liwc2015	92.0	185.0	539.0	omnibus	2.423	0.297751	0.003		0.491876	False
focuspast_liwc2015	92.0	185.0	539.0	omnibus	2.4878	0.288254	0.0031		0.491876	False
focusfuture_liwc2015	92.0	185.0	539.0	omnibus	2.3924	0.30234	0.0029		0.491876	False
focusfuture_liwc22	92.0	185.0	539.0	omnibus	2.4431	0.294772	0.003		0.491876	False
focuspast_liwc22	92.0	185.0	539.0	omnibus	2.3901	0.302693	0.0029		0.491876	False
prep_liwc22	92.0	185.0	539.0	omnibus	2.2944	0.317525	0.0028		0.506926	False
nonflu_liwc22	92.0	185.0	539.0	omnibus	2.1424	0.342598	0.0026		0.511088	False
nonflu_liwc2015	92.0	185.0	539.0	omnibus	2.1427	0.342554	0.0026		0.511088	False
negate_liwc22	92.0	185.0	539.0	omnibus	2.2023	0.332491	0.0027		0.511088	False
negate_liwc2015	92.0	185.0	539.0	omnibus	2.2139	0.330561	0.0027		0.511088	False
quant_liwc22	92.0	185.0	539.0	omnibus	1.9415	0.378791	0.0024		0.555967	False
quant_liwc2015	92.0	185.0	539.0	omnibus	1.795	0.407579	0.0022		0.588726	False
you_liwc22	92.0	185.0	539.0	omnibus	1.7358	0.419839	0.0021		0.596958	False
you_liwc2015	92.0	185.0	539.0	omnibus	1.7018	0.427024	0.0021		0.597833	False
anx_liwc22	92.0	185.0	539.0	omnibus	1.4169	0.492415	0.0017		0.653183	False
posemo_liwc22	92.0	185.0	539.0	omnibus	1.491	0.474501	0.0018		0.653183	False
auxverb_liwc22	92.0	185.0	539.0	omnibus	1.4053	0.495271	0.0017		0.653183	False
posemo_liwc2015	92.0	185.0	539.0	omnibus	1.458	0.482391	0.0018		0.653183	False
auxverb_liwc2015	92.0	185.0	539.0	omnibus	1.3425	0.511061	0.0016		0.66438	False
they_liwc22	92.0	185.0	539.0	omnibus	1.2839	0.526267	0.0016		0.664645	False
they_liwc2015	92.0	185.0	539.0	omnibus	1.2578	0.533177	0.0015		0.664645	False
anx_liwc2015	92.0	185.0	539.0	omnibus	1.229	0.540924	0.0015		0.665191	False
adverb_liwc2015	92.0	185.0	539.0	omnibus	1.075	0.584207	0.0013		0.708838	False
adverb_liwc22	92.0	185.0	539.0	omnibus	0.9879	0.610226	0.0012		0.730665	False
negemo_liwc22	92.0	185.0	539.0	omnibus	0.7529	0.686296	0.0009		0.800679	False
negemo_liwc2015	92.0	185.0	539.0	omnibus	0.6432	0.724991	0.0008		0.835116	False
verb_liwc22	92.0	185.0	539.0	omnibus	0.6007	0.740574	0.0007		0.842403	False
we_liwc22	92.0	185.0	539.0	omnibus	0.5711	0.751594	0.0007		0.844384	False
we_liwc2015	92.0	185.0	539.0	omnibus	0.5077	0.775819	0.0006		0.86097	False
verb_liwc2015	92.0	185.0	539.0	omnibus	0.4796	0.786782	0.0006		0.862616	False
affect_liwc2015	92.0	185.0	539.0	omnibus	0.4319	0.805793	0.0005		0.865519	False
affect_liwc22	92.0	185.0	539.0	omnibus	0.4253	0.808451	0.0005		0.865519	False
focuspresent_liwc22	92.0	185.0	539.0	omnibus	0.3995	0.818949	0.0005		0.866562	False
compare_liwc2015	92.0	185.0	539.0	omnibus	0.2867	0.866469	0.0004		0.886659	False
compare_liwc22	92.0	185.0	539.0	omnibus	0.3035	0.859187	0.0004		0.886659	False
focuspresent_liwc2015	92.0	185.0	539.0	omnibus	0.285	0.867172	0.0003		0.886659	False
sad_liwc2015	92.0	185.0	539.0	omnibus	0.0648	0.968108	0.0001		0.971362	False
sad_liwc22	92.0	185.0	539.0	omnibus	0.0581	0.971362	0.0001		0.971362	False